SPECIFICATION UX/UI & UML

# CIVORA Architecture Specification

AI-Assisted Structural Engineering Platform — UX/UI, UML & Software System Design

## 1. System Overview

Civora is an AI-assisted structural engineering platform that converts architectural CAD/Revit drawings into an analyzed, designed, and documented reinforced-concrete structural system.

**Architectural Input → Drawing Ingestion → AI Understanding → Structural Model → Load Gen → FEA Analysis → Design → Review → Documents**

## 2. Core Design Principle

**AI proposes → Engineering engine calculates → Engineer reviews → Engineer approves → System generates.**

Civora is engineered as an interactive workspace rather than a black box. At every stage, the engineer inspects detections, evaluates confidence scores, edits parameters, locks approved elements, reruns calculations, and monitors discrepancies.

## 3. Main Application Modules

### Core & Data Modules

- Authentication & User Management
- Project & Drawing Management
- AI Drawing Recognition Engine
- Project / Site Information
- Structural Canonical Model

### Engineering & Output Modules

- Load & Analysis (FEA) Engine
- Design Engine & Member Grouping
- Reinforcement & Detailing Database
- Drawing & Document Generators
- Review, Approval & BOQ Engines

## 4. Project Creation & Dashboard

Tracks project name, unique ID, client, location, engineer credentials, project type, floors, active code books, status, creation date, and last modified timestamps. Visual status badges track lifecycle progression from initial upload to final approval.

# Drawing Ingestion & Intelligence

Input processing, title block extraction, and site parameter generation

## 5. Drawing Input Module

**Supported Formats:** DWG, DXF, RVT, PDF, and raster images. Future expansion includes IFC and interoperable BIM formats.

**Upload Drawing → File Validation → Parser → Layer/Object Extraction → OCR → Geometry → AI Recognition → Engineering Data Model**

*Note:* The original source drawing is always maintained separately from the interpreted engineering model.

## 6. Drawing Intelligence Engine

Automated title block detection extracts critical metadata to streamline configuration:

### Extracted Metadata

- Project Name, Location & Client
- Drawing Number & Revision Index
- Drawing Scale & Date
- Lead Architect Information

### Site Integration

The detected project location serves as the primary automated input parameter for the Site Information Engine, driving geographic database queries.

## 7. Site Information Engine

Queries regional databases to compile site criteria, clearly distinguishing AI-suggested defaults from engineer-confirmed values with complete override permissions.

- **Soil Parameters:** SBC & allowable bearing capacity.
- **Environmental Actions:** Design wind speed and exposure categories.
- **Seismic Data:** Acceleration coefficients and zone parameters.

## 8. Code Selection

Modular code selection controls load calculations, load combinations, concrete design parameters, reinforcement ratios, wind/seismic criteria, and serviceability detailing rules across international standards.

# Structural Elements & Canonical Model

Element detection, level hierarchy, and single source of truth architecture

## 9. Architectural Plan & Grid Detection

Extracts boundaries, rooms, walls, doors, windows, stairs, cores, shafts, openings, and grid lines. Establishes a verified coordinate mapping between drawing units and real-world dimensions with complete manual controls (rename, move, add, delete, lock).

## 10. Structural Element Detection

### Columns & Beams

**Columns:** Evaluated at grid intersections, wall junctions, and building corners with confidence scoring.
**Beams:** Generated from connectivity, spans, and wall alignments.

### Slabs & Levels

**Slabs:** Panel identification based on structural framing and boundary geometry.
**Levels:** Ground, foundation, basement, floor-to-floor heights, and roof levels.

## 11. Canonical Structural Model (Single Source of Truth)

The centralized internal data representation shared downstream by analysis, design, and drawing generators, preventing data inconsistencies:

```
PROJECT → [ Site | Code | Materials | Levels | Grids | Nodes ]
ELEMENTS → [ Columns | Beams | Slabs | Walls | Foundations ]
ANALYSIS → [ Loads | Load Cases | Load Combinations | Analysis Model ]
```

# Analysis, Design & Grouping

FEA processing, member design rules, and automated classification

## 12. Load & FEA Analysis Engine

Calculates dead loads (self-weight, finishes, partitions), live loads based on occupancy, and environmental wind/seismic actions. The FEA preprocessor builds stiffness matrices and boundary conditions to solve for displacements, axial forces, shear, bending moments, and torsion.

## 13. Design Engine & Element Modules

### Beam & Column Design

**Beams:** Evaluates span, Mu, Vu, Tu, deflection, and required reinforcement.
**Columns:** Evaluates axial-flexure interaction, biaxial effects, slenderness, and second-order behavior.

### Slab & Foundation Design

**Slabs:** Analyzes panel dimensions, moments, and serviceability.
**Foundations:** Supports isolated, combined, strip, raft, and pile caps based on soil pressures and reactions.

## 14. Member Grouping Engine

Classifies similar structural members post-design using parameter normalization and clustering algorithms rather than isolated forces, optimizing construction efficiency (e.g., Beams B1/B2, Columns C1/C2).

# Drawings, Documents & Traceability

Automated documentation generation, version control, and change propagation

## 15. Drawing & Document Generator

Receives verified engineering data to produce structural layouts, schedules, detailing notes, calculation reports, BOQs, and material schedules across PDF, DWG, DXF, RVT, and Excel formats.

## 16. Confidence System & Approval Workflow

AI detections expose clear confidence percentages (e.g., Grid A-3: 99.7%, Column C5: 96.2%). Low-confidence items trigger mandatory engineering review states: *Detected, AI Suggested, Modified, Approved, Rejected, Locked,* or *Needs Review*.

## 17. Critical Traceability Chain & Change Propagation

Maintains end-to-end traceability across the asset lifecycle:

```
Grid B-4 → Column C17 → FEA Element 183 → Design Result → Group C3 → Reinforcement Detail
```

**Change Propagation:** Modifying an element (e.g., Column C17 section change) automatically flags downstream connected beams, analysis models, design results, and drawing sheets as invalid, prompting an update-and-rerun action.

# UML Architecture & System Design Brief

Core class model structure and UI workspace framework

## 18. UML Deliverables & Class Model

The platform architecture models system context, use cases, components, classes, sequences, activity, state machines, entity relationships, and deployment modules.

```
CORE CLASSES → [ User | Project | Drawing | Site | CodeBook | Material | Level | Grid | Node ]
ELEMENT CLASSES → [ Column | Beam | Slab | Foundation | Load | AnalysisResult | DesignResult ]
OUTPUT CLASSES → [ Reinforcement | MemberGroup | Document | Approval ]
```

## 19. Main UX Workspace Structure

### Workspace Layout

Features top-level action menus (Project, Code, Save, Run, Review, Export), a left-hand project tree, a central 2D/3D model viewport, and a right-hand property inspector.

### Designer Brief

Design Civora as an integrated engineering system. Make the AI-to-engineer workflow fully transparent, support robust 3D/2D interaction, and streamline review workflows.